CAREER GUIDANCE FOR NON-TRADITIONAL STUDENTS

25-26J-336

Project Proposal Report

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B.Sc. (Hons) in Information Technology Specializing in Information Technology

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DECLARATION

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We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text

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ABSTRACT

Non-traditional and marginalized learners often face substantial barriers in navigating effective education-to-employment pathways. There is an urgent need for alternate pathways to skill acquisition and job security because a sizable fraction of young people in Sri Lanka and around the world seek education outside of the state university system. For Sri Lanka's non-traditional learners, such as those who choose non-university pathways or lack Advanced Level credentials, this study suggests designing and implementing an AI-powered Learn-to-Earn Diagram Generator. Large language models (LLMs) and survey-based user profiling are combined by the system to create customized learning-to-career pathways, which are illustrated by interactive visual diagrams that link various educational phases to related career outcomes. Learners can learn about the time and cost value of chosen pathways using an integrated ROI estimator that uses local labour market data, and recommendations are dynamically adjusted by an iterative feedback loop and milestone-tracking module to maintain engagement. The proposed system aims to improve clarity, motivation, and actionable planning among marginalized learners by fusing visual scaffolding with AI-driven personalization. The focus of the preliminary assessment is on the pathways' interpretability, usability, and quantifiable influence on learners' confidence and goal setting. Finally, by providing an approachable, goal-oriented, and expandable guidance tool, this study fills in structural gaps in Sri Lanka's education-employment ecosystem and may be a model that can be applied to other developing nations.

ABBREVIATIONS

Abbreviation	Description
ROI	Return on investment
AI	Artificial Intelligence
ILO	International Labour Organization
LLM	Large Language Model

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1. INTRODUCTION

A significant percentage of the global workforce-in-waiting are non-traditional learners, or people who do not pursue traditional university pathways. Despite formal educational achievements, youth in low- and middle-income countries often lack access to skill-aligned education and structured career guidance, which leads to high rates of underemployment and unemployment, according to the International Labour Organization (ILO)[1]. Only 15–17% of Sri Lanka's 250,000 students who take the GCE Advanced Level (A/L) exam each year are accepted into public universities[2]. Without consistent, individualized guidance, the remaining students must navigate a complex web of private, informal, and vocational learning opportunities[3].

Educational technology and career advice research has investigated recommender systems, intelligent tutoring systems, and AI-powered counselling aids. This project's relevant techniques include LLMs for natural language interaction[4], recommendation systems for personalized courses and career matching[5], labour market analytics for ROI estimation[6], and visual scaffolding methods for educational pathway representation[7]. Mastery of these strategies is required while developing an integrated guidance system.

Modern solutions, such as Jobiri, an AI-powered digital career advisor, provide scalable, personalized coaching in European contexts[8]. Coursera's AI Coach and Google Career Certificates combine course suggestions and employment pathways, resulting in promising career results[9], [10]. However, these tools are largely built for industrialized markets and do not consider Sri Lanka's socioeconomic, linguistic, and sectoral characteristics.

Previous attempts in Sri Lanka have included government-led career advisory units and donor-funded employability programs[3],[11]. but they are still limited in scale, personalization, and integration with real-time labour market data. The proposed Learn-to-Earn Diagram Generator expands on international best practices by incorporating context-aware personalization, bilingual accessibility, and ROI-driven pathway visualization customized to Sri Lanka's educational-employment ecosystem. Unlike

previous systems, it features a feedback-driven milestone tracker, allowing for iterative development of recommendations and adaptation to changing labour market demands.

2. BACKGROUND

In many low and middle-income nations, including Sri Lanka, non-traditional learners those who do not follow established university pathways continue facing limitations to structured, personalized career coaching. In accordance with the International Labour Organization (ILO), despite having a high level of education, a significant percentage of young people in these situations do not have clear routes to employment, which contributes to high unemployment and underemployment rates[1]. Only 15–17% of the 250,000 students who take the GCE Advanced Level (A/L) exam each year in Sri Lanka is accepted into public universities[2]. The others, frequently without trustworthy, current guidance, must negotiate a disjointed ecology of private institutions, informal learning possibilities, and vocational training centers[3].

There is a lot of evidence that educational outcomes and labour market needs don't match up. A YouLead report from 2020 says that "skills, aspirational, and information gaps" are primary reasons why there are both unemployed young people and jobs that aren't being filled in important industries[7]. Career advice services in Sri Lanka, including those in the general education sector, are limited in scope, lack professional training for counsellors, and are rarely connected with labour market analytics[3]. To close this gap, we need systems that can grow and adapt to different situations and provide personalized, evidence-based paths from learning to earning.

3. LITERATURE SURVEY

A. AI-Driven Career Guidance Systems

Recent progress in AI has made it possible to create personalized career coaching systems that can be used by many people at once. Jobiri, an AI-based digital career advisor, providing on-demand assistance, resume optimization, and job matching in European contexts[6]. Coursera's AI Coach uses LLMs to suggest role-specific learning routes based on what the user wants to learn. The company says this has led to better engagement and learning outcomes[12]. Google Career Certificates offer short-term training and AI-powered job matching. More than 70% of graduates say they have had good career outcomes within six months[10]. These tools show how AI can make employment advice more accessible, personalized, and relevant. But they are mostly made for developed markets and don't work well in Sri Lanka.

B. Recommendation Systems and Learning Path Generation

Traditional course recommendation systems involve collaborative filtering and content-based algorithms[13], but these approaches generally suffer from the “cold start” problem and limited personalization[12]. LLM-powered solutions have emerged to solve these limitations by analyzing natural language inputs and mapping them into semantically appropriate courses[5]. GPT-4-based learning path generator that dynamically modifies based on learner success, combining multimodal data for individualized pathway construction[7]. Such models exhibit considerable promise for adapting to varied learner profiles, but their applicability in marginalized or resource-limited environments remains underexplored.

C. ROI Estimation in Educational Guidance

Outcome-focused advice increasingly incorporates return-on-investment (ROI) measures to inform learner decision-making. Coursera’s 2023 research found that 77% of learners globally reported professional gains from online courses, with substantially greater rates in underdeveloped countries[6]. Google Career Certificates similarly track job and wage improvements to validate training efficacy[5]. Integrating

ROI estimation into learning paths might enable under-represented learners often constrained by financial and time resources prioritize high-impact training opportunities.

D. Gaps in Existing Literature and Practice

While worldwide platforms offer advanced guiding solutions, Sri Lanka lacks customized, AI-driven systems that combine career recommendations, visual pathway development, and ROI metrics. Government-led career advising activities[3] to have limited reach, lack personalization, and are separated from real-time labor market data. No published studies have yet evaluated the use of LLMs for bilingual, context-aware career pathway development in Sri Lanka, reflecting a clear research and implementation gap that this project attempts to address.

4. RESEARCH GAP

There is still an obvious lack of locally tailored solutions for non-traditional students in Sri Lanka, despite recent developments in AI-driven career counselling showing promise for scalable and customized education-to-employment pathways. Current global platforms like Google Career Certificates, Jobiri, and Coursera's AI Coach are largely made for developed economies and rely on labour market structures, datasets, and pedagogical frameworks that don't consider the socioeconomic realities, linguistic diversity, and sectoral employment demands of Sri Lanka.

Existing career guidance programs in Sri Lanka, like those included in the general education system, are rarely integrated with real-time labour market intelligence, have a limited range, and lack qualified counsellors. Even when there are open positions in priority sectors, persistent "skills, aspirational, and information gaps" are a contributing factor to high youth unemployment, according to national and donor reports. This suggests a structural failure to link the goals of learners with the changing demands of the labour market.

Furthermore, the use of LLM to create bilingual, context-aware, and ROI-informed career pathways for marginalized or vocational learners has not been studied in any previous research. There is a big gap in comprehensive, feedback-driven pathway generation that supports vocational, external, and non-university students because previous AI-based recommender systems mostly focus on single-course recommendations or learners who are university-focused.

By combining ROI estimation, localized labour market intelligence, LLM-powered personalization, and culturally aware user experience design into a single Learn-to-Earn Diagram Generator, this proposal seeks to close this gap. By doing this, it will offer a fresh, empirically supported framework that directly benefits Sri Lanka's non-traditional learners and serves as a model that can be replicated in other emerging contexts.

Feature	Jobiri (AI Coach)	Coursera Coach	Google Certs	Traditional E-Learning	National Services	Learn-to-Earn
Personalization (AI-driven)	🟢	🟡	🟡	🔴	🟡	🟢
ROI Estimation for Pathways	🟡	🔴	🟡	🔴	🟡	🟢
Pathway Diagram Generation	🔴	🔴	🔴	🔴	🔴	🟢
Local Labor Market Integration	🟢	🟡	🟡	🔴	🟢	🟢
Milestone Tracking	🟡	🟡	🟡	🔴	🟡	🟢
Feedback Loops & Adaptivity	🟡	🟡	🔴	🔴	🟡	🟢
Target Audience Focus	🟡	🟡	🔴	🟡	🟢	🟢
		🟢 - Yes	🟡 - Partial	🔴 - No		

Figure 1: Research Gap Analysis

5. RESEARCH PROBLEM

Despite the global advancements in AI-driven career advising platforms, Sri Lankan non-traditional learners especially those who have not completed Advanced Levels or who pursue alternative educational pathways continue to face a severe lack of locally relevant and contextually aware solutions. Although they offer scalable guidance, existing international platforms like Jobiri, Coursera's AI Coach, and Google Career Certificates are primarily made for high-income economies and cannot be adjusted to Sri Lanka's socioeconomic circumstances, linguistic diversity, or labour market dynamics. The effectiveness of career guidance services in addressing youth unemployment and skills mismatches is limited locally due to their fragmentation, lack of resources, and disconnection from real-time labour market data.

Research Problem Statement: How can an AI-powered system be designed to generate personalized, bilingual, and ROI-informed "Learn-to-Earn" pathways visualized as interactive diagrams for non-traditional learners in Sri Lanka, to bridge the gap between education and employment while ensuring scalability, inclusivity, and trustworthiness?

The following sub-challenges are included in this research problem:

1. Integration of LLMs with local data: To create contextually relevant pathways, large language models are combined with course, vocational, and labour market datasets unique to Sri Lanka.
2. Creating understandable visual representations of intricate learning-to-employment pathways to improve accessibility and comprehension is known as visual scaffolding.
3. ROI estimation: Incorporating return-on-investment computations that enable students to assess the worth of educational expenditures considering time and budgetary constraints.
4. Feedback and flexibility: Putting in place feedback loops and milestone tracking to allow for ongoing enhancements to suggestions and student involvement.
5. Resolving this issue is crucial to closing long-standing gaps in Sri Lanka's education-to-employment ecosystem and creating a replicable model for other developing nations dealing with comparable issues.

6. OBJECTIVE

6.1. Main Objectives

To improve non-traditional learners' capacity to make well-informed decisions about their education and careers, an AI-powered Learn-to-Earn Diagram Generator that offers individualized, bilingual, and ROI-driven learning-to-career pathways are being designed, developed, and evaluated.

SMART Alignment:

- Specific: Concentrates on developing an AI system for creating pathways that are suited to Sri Lanka's non-traditional students.
- Measurable: Improvements in learner outcomes, system accuracy, and usability are used to gauge success.
- Achievable: Doable given the EdTech infrastructure already in place, survey-based data collection, and available LLMs.
- Realistic: Complies with technological capabilities and local needs (high youth unemployment, limited guidance services).
- Time-bound: The goal is to finish the research within 12 months.

6.2. Specific Objectives

I. Data Collection

- This goal is specific to collecting learner profiles from four target groups; it is measurable with at least 3,000 records; it can be accomplished through collaborations with institutions; it is realistic because stratified sampling guarantees representativeness.

II. System Design

- This goal is specific to creating modules for input, evaluation, suggestions, diagram creation, ROI estimation, and tracking; it is quantifiable by producing a fully functional prototype; it is feasible with current frameworks; it is reasonable considering the resources at hand.

III. Model Development

- This goal is specific to create ML and LLM-based models for pathways, recommendations, and return on investment. It is measurable using metrics, achievable with standard algorithms and cloud support, realistic given the infrastructure and dataset.

IV. Implementation

- Achievable with Next.js, APIs, and NoSQL, this goal is specifically deploying a web application with interactive diagrams and milestone tracking. It is realistic because it supports common devices, time-bound with a 12-month deployment deadline, and measurable and successful user task completion.

V. **Evaluation**

- This goal is specific to evaluating usability, clarity, and user impact; it is measurable by a 15% increase in program enrolment and positive learner feedback; it can be accomplished through pilot studies and tracking; it is realistic in accordance with EdTech evaluation practices.

7. METHODOLOGY

The AI-driven Learn-to-Earn Diagram Generator study methodology consists of five parts: data collection, system design, model development, implementation, and evaluation.

1. Data Collection

To build the course prediction model, participants will be selected from four learner groups in Sri Lanka:

a. Students from private institutes
b. Students on international scholarships
c. Students from external colleges and universities
d. Students from vocational training centers

Stratified sampling will guarantee variation across districts, genders, age groups, and languages. For model training and validation, the target sample size is at least 3,000 learners. There should also be an extra group of learners who are not being tested yet.

- **Data Content**: Age, Language, Preferred language, Education level, O/L results, A/L results (optional), IELTS (optional), Area of interest, Income each month, When you're available (on weekends or weekdays), Target completion period, Location, Preferred places, Study method (online/onsite/hybrid), Gender.
- **Annotation**: Actual course enrolments, expert validation, and learner outcomes (employment, salary, completion) will be used to generate ground truth labels.
- **Preprocessing**: Data cleaning, normalization, missing value imputation, categorical encoding, balancing class distributions, and feature scaling.
- **Privacy concerns**: We won't collect any personally identifiable information, such names, addresses, or national IDs. All records will be anonymized and encrypted to comply with ethical standards.

2. System Design

The system will be designed with three operational phases and the following modules:

- Input Module: A survey form to record user preferences and constraints in real time.
- Assessment Model: Analyzes user profile to identify eligibility, limits, and readiness levels.
- Personalization Module: Generates a sorted list of acceptable programs considering user preferences, budget, and availability.
- Recommendation Engine: Ranks programs using appropriateness, cost, and ROI metrics.
- Diagram Generator: Produces a personalized “Learn-to-Earn” diagram relating learning processes to professional outcomes.
- ROI Estimator: Predicts time-based and financial return on investment for each recommended pathway.

- Milestone & Feedback Module: Creates individual milestones and records user feedback for iterative improvement.

3. **Model Development**
    - Prediction Model: A trained to recommend suitable courses/programs.
    - Language Model Integration: GPT-4 or equivalent LLM will be integrated to generate pathway explanations, diagrams, and career guidance.
    - Feature Engineering: Encodes categorical variables and scales numeric variables.
    - Training & Validation: Dataset split into training, validation, and testing sets with stratification to ensure balanced representation.
    - Evaluation Metrics: Mean Absolute Error (MAE) for ROI prediction.

4. **Implementation**

    A fully interactive web-based application will be developed with the following features:
    - Frontend: Web application interface for survey input, visualization of diagrams, ROI graphs, and milestone tracking.
    - Backend: RESTful APIs handling data storage, model inference, and recommendation processing.
    - Database: NoSQL for structured learner data and program metadata.
    - Integration: LLM APIs for diagram and text generation; analytics dashboard for monitoring outcomes.
    - Security: Authentication, role-based access control, and encrypted data storage to ensure confidentiality.

5. **Evaluation**

    A combination of user-centric and longitudinal assessments will be used to evaluate the suggested system. A subset of students will participate in a pilot study to assess the interface's usability, the clarity of the diagrams produced, and the

perceived value of the suggestions using validated tools and the System Usability Scale. The effectiveness of the system in enhancing education-to-employment transitions will also be evaluated through a longitudinal tracking of learner outcomes, such as program enrolment, milestone completion, and employment status. The results will be compared to baseline groups.

System Overview

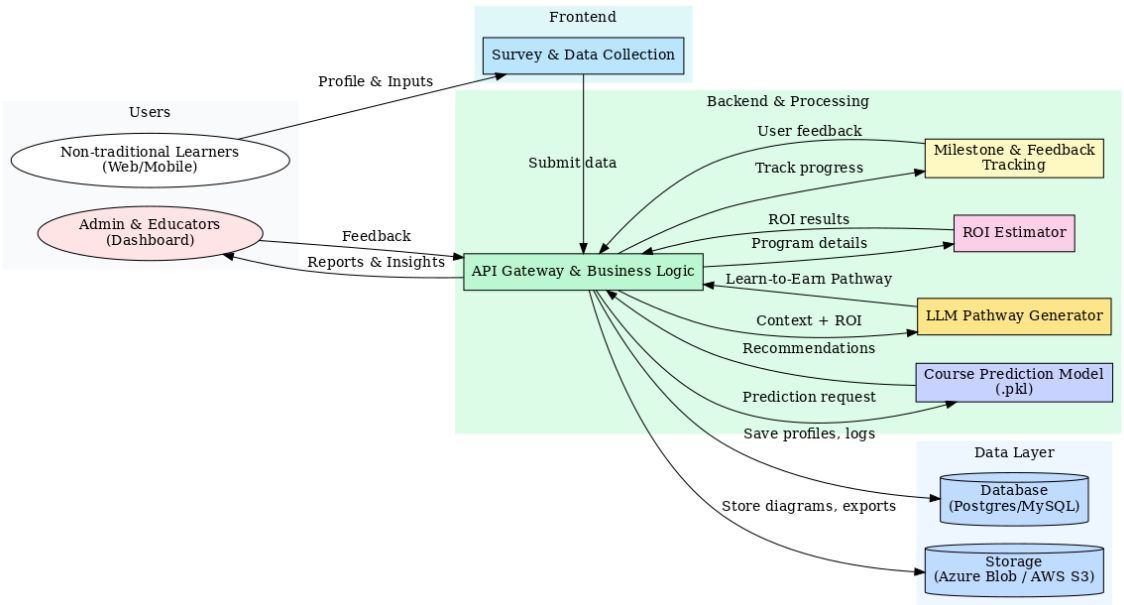

Figure 2: System Overview

8. PROJECT REQUIREMENTS

8.1. Functional Requirements

This research system's primary functional requirements are:

- **Survey-Based Data Collection**: A structured survey will be used by the system to gather user data, including demographics, educational background, preferences, and limitations.
- **Course Prediction and Ranking**: Using user profiles, preferences, and eligibility requirements, the system will forecast and rank appropriate programs or courses.

- **Personalized Pathway Generation**: A Learn-to-Earn diagram that links suggested learning activities to career outcomes will be produced by an integrated LLM module.
- **ROI Estimation**: Program cost, duration, and anticipated earning potential are all considered by the system when calculating the expected return on investment (ROI).
- **Milestone Creation and Tracking**: Users will be able to create custom milestones, mark them as finished, and share updates on their progress.
- **Feedback Loop**: To enhance suggestions and gradually improve pathway creation, user input will be gathered.
- **Visualization**: To represent pathways and estimate ROI, the system will produce interactive visual diagrams and graphs.

8.2. Non-Functional Requirements

- **Performance**: After receiving user input, the system must produce pathway diagrams and recommendations in small periods.
- **Scalability**: Thousands of concurrent users across districts must be supported by the platform without causing a drop in service quality.
- **Security and Privacy**: All user information must be anonymized and kept in a secure location while adhering to data protection regulations. No names or national identification numbers, or other personally identifiable information, will be gathered.
- **Usability**: Students with different levels of digital literacy should be able to easily navigate and use the interface.
- **Reliability**: The system must be able to recover gracefully from crashes or interruptions and maintain an up-time of at least 99%.
- **Maintainability**: For upcoming updates and integration with new models, the codebase should be easily extensible, modular, and well-documented.

- **Accessibility**: To ensure that it is available to both rural and urban learners, the system should support typical household devices with internet access.

8.3. System Requirements

- **Hardware Requirements:**
  - Devices: A laptop or desktop computer with internet access; a tablet or smartphone for on-the-go use.
  - Storage: Logs, application files, and datasets require at least 10 GB.
  - Memory: For seamless client-side and backend operations, a minimum of 16 GB of RAM is advised.
- **Software Requirements:**
  - Operating Systems: Cross-platform support (Windows, macOS, Linux, Android, iOS).
  - Programming Frameworks:
    - TensorFlow/PyTorch for creating models.
    - scikit-learn/LightGBM for ROI regression and prediction models.
    - For front-end development, use React/Next.js.
    - Express.js and FastAPI are used for backend services.
    - PostgreSQL is used to store structured learner data.
    - libraries for graph rendering for visualizing diagrams.
    - OpenAI API (GPT-4 or comparable LLM) for creating diagrams and explaining pathways.
  - Cloud Services:
    - Use of Microsoft Azure for scalable computing, storage, and security administration.

8.4. User Requirements

- Get recommendations that are specific to their academic background, preferences, and limitations.

- Determine clear, actionable Learn-to-Earn charts and ROI projections to help guide choices.
- Utilize feedback features to keep track of personal milestones and evaluate your progress.
- Interact with the system's preferred language (English)

8.5. GANTT Chart

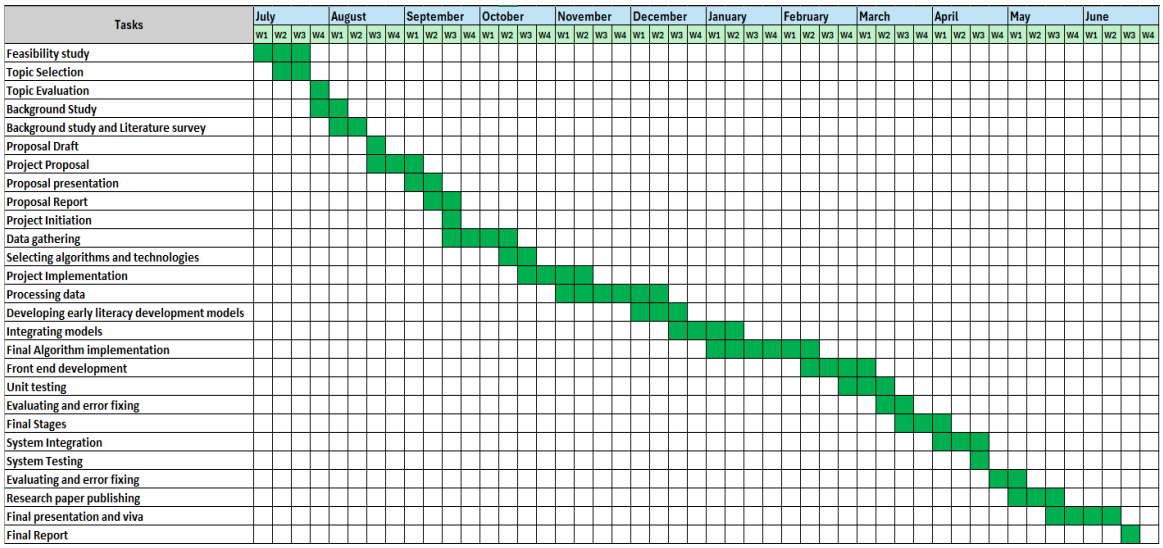

Figure 3: Gantt Chart

8.6. Work Breakdown Structure

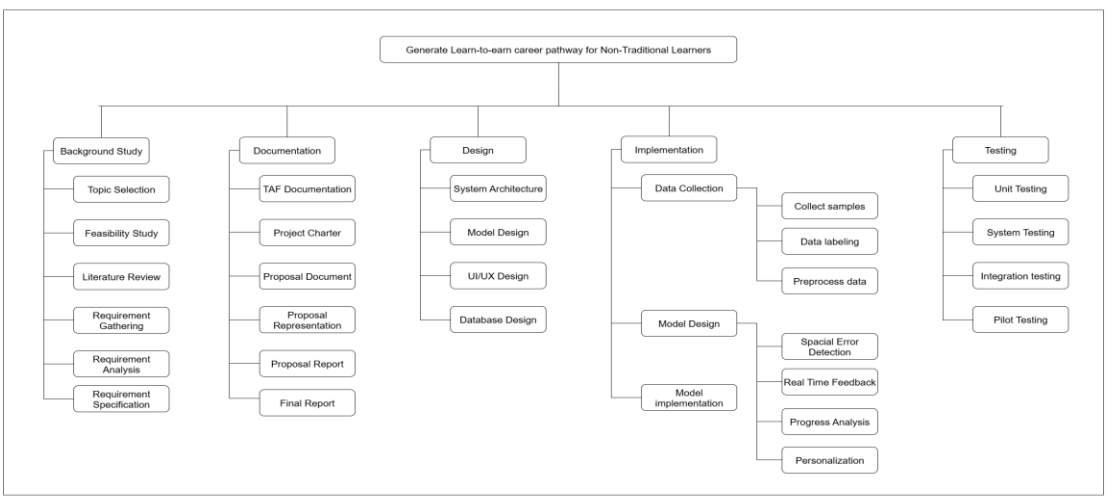

Figure 4: Work Breakdown Structure

9. BUDGET ALLOCATION

- **Expert Consultation**: To guarantee correctness, inclusivity, and contextual alignment of learning-to-employment pathways, education specialists, vocational training advisors, and labour market experts are consulted.
- **Data collection and annotation**: Include surveying participants, preprocessing and annotating learner profiles to produce a strong training dataset, and recruiting participants from private institutions, vocational centers, scholarship programs, and outside universities.
- **Software, Tools, and Subscriptions**: Database systems, cloud computing resources, Glassdoor API for insights into salaries and labour markets, GPT subscription for model integration and fine-tuning, and licensing fees for AI frameworks (TensorFlow, PyTorch).
- **System Development & Integration**: Costs associated with the creation and deployment of the ROI estimator, input module, personalization engine, diagram generator, and milestone-tracking dashboard.
- **Testing and validation**: evaluating the system's usability, the clarity of the generated diagrams, and the precision of ROI predictions through user trials with various learner groups.
- **Operational Costs**: Paying for long-term system maintenance, technical assistance, documentation, security compliance, and administrative costs.

10. CONCLUSION

This study suggests creating a Learn-to-Earn Diagram Generator with AI capabilities specifically for non-traditional Sri Lankan students who are frequently shut out of conventional university programs. Through the integration of large language models, machine learning, and survey-based user profiling, the system seeks to produce interactive visual diagrams that illustrate individualized educational-to-employment pathways. Iterative feedback loops, milestone tracking, bilingual accessibility, and ROI estimation are some of the key features that set this system apart from other local and international solutions.

The lack of context-aware career counselling, the limited personalization of current tools, and the lack of integration with real-time labour market data are just a few of the significant issues in Sri Lanka's education–employment ecosystem that are addressed in this study. The suggested approach boosts learner confidence, motivation, and employability while also improving clarity and decision-making by placing students within a clear roadmap that links study choices to career outcomes.

In addition to providing a scalable, goal-oriented workforce development solution, this work advances the academic domains of educational technology, AI-driven personalization, and career pathway visualization. Outside of Sri Lanka, the concept might be helpful in other low- and middle-income environments with similar structural barriers to employment and higher education. This allows underprivileged students to seek equitable and sustainable professions by bridging the gap between school and employment. It helps with research as well.

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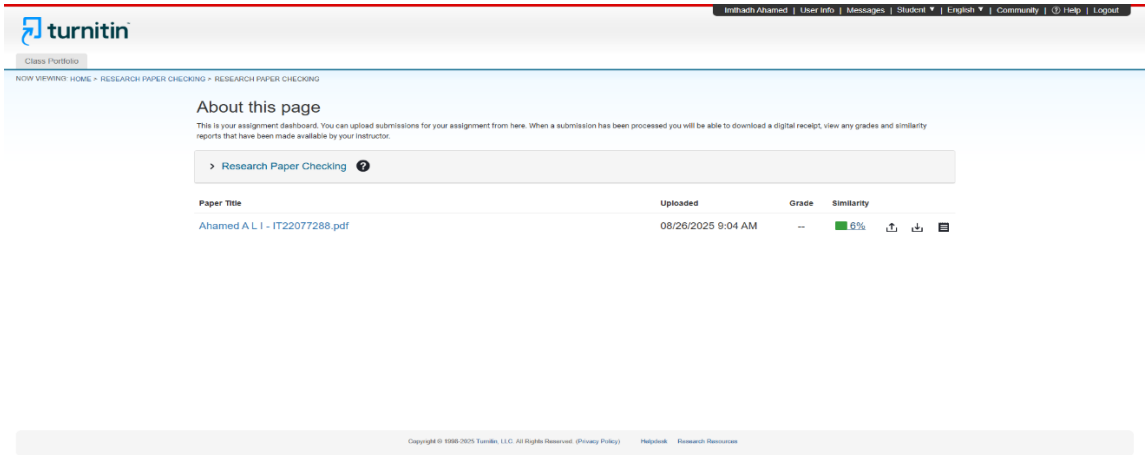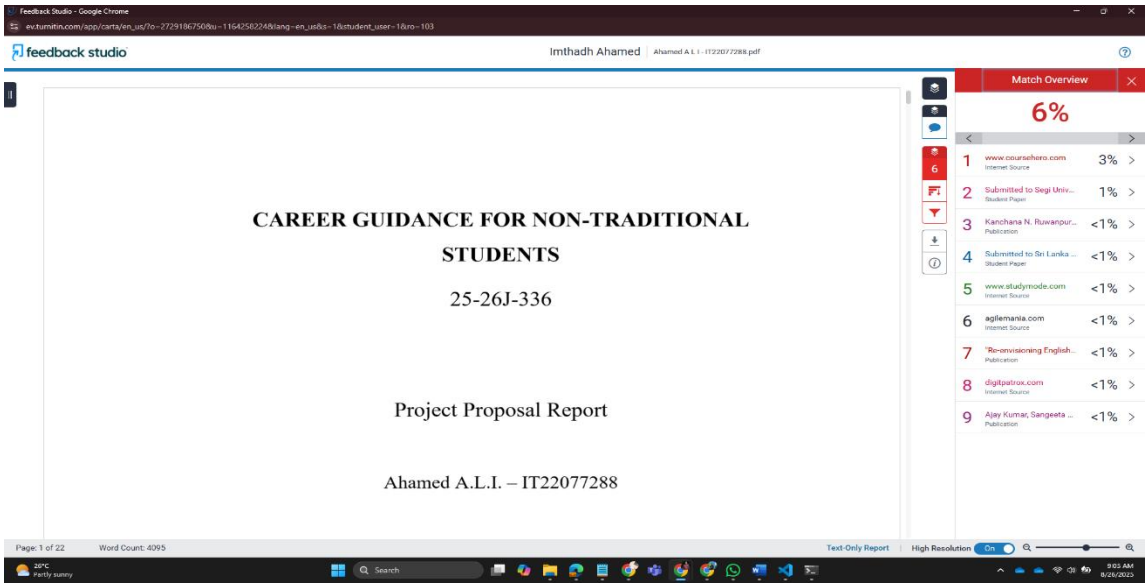
12. APPENDIX